

Scalable Large Model for Unlabeled Anomaly Detection With Trio-Attention U-Transformer and Manifold-Learning Siamese Discriminator

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Abstract—To identify pattern deviations in large-scale industrial infrastructures, anomaly detection is crucial yet challenging. Previous research has not adequately addressed the characteristics and deployment considerations in these complex scenarios. In this paper, we present *InoU*, a scalable anomaly detection framework to process unlabeled multivariate time-series data. We incorporate a VAE filter to ease impacts from noisy components in training materials. We propose a scalable trio-attention U-Transformer to construct the typical representation of high-dimensional streams and produce pseudo labels that enable the later training process. The ultra perception and intra-/ inter-flow attention mechanisms are delicately designed to aggregate information from different flows with variable granularities while keeping a global view of the data. Its nested structure helps to maintain high efficiency even when the model is scaled down. We introduce a Siamese discriminator that projects target data into manifolds, and collates discrepancies at the embedding level. This paradigm elevates detection performance far beyond segment-wise error comparison in prior works. We apply contrastive and adversarial learning techniques to optimize manifold projection and detection performance when processing unseen samples. Extensive experiments on five large-scale datasets demonstrate the effectiveness of *InoU* with an average *F1-Score* improvement of 5.58%, significantly outperforming the state-of-the-art.

Index Terms—Anomaly detection, service and system management, self-supervised, Siamese network, Transformer, contrastive, manifold learning.

I. INTRODUCTION

THE massive deployment of sensors, actuators, and other assets in industrial scenarios has formed an increasingly complex network architecture [1], [2], enabling a series of automated and intelligent operations [3], [4]. The deployments, in turn, pose pressing urges for their maintenance [5]. Anomaly detection, which identifies unusual or unexpected patterns or

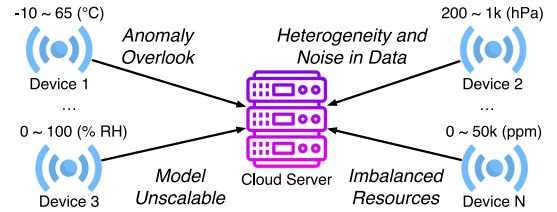


Fig. 1. Challenges of deploying anomaly detection in industrial scenarios.

events in the target data, provides a new viewpoint to address this need [6], [7], [8], thus attracting widespread attention.

Most existing solutions [9], [10], [11], [12], [13] use data normalization and windowing techniques to pre-process the data, and deploy generative models to extract patterns and reconstruct the target data. The criteria for identifying anomalies are usually based on the segment-wise reconstruction error of the normalized data. Seemingly promising, these solutions face challenges (Fig. 1) caused by a series of factors in actual industrial deployment, including the massive volume and variety of on-site data, the heterogeneity in different devices, and the imbalanced hardware resources in different configurations.

First, the insufficient consideration of discrepancy analysis in prior works contributes to the potential neglects of anomalies. Differences in high-dimensional data easily become subtle after segment-wise averaging, which is a common practice of error calculation. Moreover, the data normalization that transforms data with variable ranges to a uniform scale, potentially contributes to overlooking fluctuations in channels with more dynamic ranges.

Second, the choice of window length and the inadequate availability of labeled data further cause barriers in actual industrial practice. Longer windows dilute the presence of data points that deviate from the norm, while shorter windows fail to capture sufficient contextual information over time. Moreover, the labeling process in industrial scenarios is costly and labor-intensive, while the model tuning process in a self-supervised manner has not yet been fully explored.

Lastly, the resources in industrial scenarios are usually imbalanced, posing challenges to the scalability of models to be deployed. While Multi-Access Edge Computing (MEC) offers a potential solution for near-source model inference, its

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computational capabilities are inherently limited. Resources on clouds are usually more abundant, but remote deployment is not acceptable for latency-sensitive applications. These characteristics of a typical industrial environment determine that the anomaly detection framework should maintain adaptability and scalability.

To address these challenges, we redesign an anomaly detection pipeline that best fits industrial needs, and propose a scalable framework *InoU* that incorporates a trio-attention U-Transformer and a manifold-learning Siamese discriminator under a unified yet flexible architecture with tunable parameters ranging from approximately 100 million to 6 billion, depending on the specific configuration and data source. During the design of its architecture, we adopt a complementary rationale, where the unified structure utilizes generative models to extract typical patterns and reconstruct data, while leveraging discriminative models for manifold projection and discrepancy analysis.

Adapting to the data properties under industrial setups, we decouple the VAE (Variational Autoencoder) model from direct involvement in the anomaly detection task, but to filter irregularities and noise from the raw training materials. In this way, we bypass the risk of the VAE model being biased or improperly trained, and offset the impacts of noise and anomalies without the need for human intervention.

To learn the embeddings of normal data while maintaining a high level of scalability, we propose a trio-attention U-Transformer. The ultra perception and intra-/inter-flow attention mechanisms within each flow assist information aggregation in variable granularities without losing a high-level perspective of the complex industrial data. The U-shaped nested structure also helps the model maintain high efficiency even when the model is scaled down.

Furthermore, we devise a contrastive manifold-learning Siamese discriminator to cope with the challenge of anomaly overlooking in conventional numerical error comparison based works while processing high-dimensional industrial data. Furthermore, we design a contrastive manifold-learning Siamese discriminator to address the limitation of conventional numerical error comparison methods, which often encounter the challenge of anomaly overlooking when processing high-dimensional industrial data. This structure projects the observed and reconstructed sequences into the manifold through a shared backbone, and adaptively collates their discrepancies at the embedding level. We introduce techniques from contrastive and adversarial learning to the duet outputs from the Siamese backbones to optimize the manifold projection approach and achieve a more constructed latent space.

The contributions of this work are summarized here.

- We present the anomaly detection framework *InoU*, incorporating a trio-attention scalable U-Transformer reconstructor and a manifold-learning Siamese discriminator in a unified architecture. This design inspects embedding discrepancies in a fine-grained manner while maintaining high scalability.
- We design a refined data processing pipeline to offset the impacts of noise and anomalies in the raw training materials and adapt to industrial properties. We also

introduce contrastive and adversarial learning to optimize the projection of manifold and further elevate the anomaly detection performance.

- We evaluate *InoU* on five large-scale datasets from real-world industrial applications. *InoU* achieves an overall *F1-Score* of 0.9168, outperforming the latest state-of-the-art. Extensive experiments validate its scalability and effectiveness.

The rest of this paper is organized in the following ways. Section II introduces the underlying methodology, and we further present the design details of *InoU* in Section III. In Section IV, we discuss the settings and results of the extensive experiments. Section V briefly reviews recent research on multivariate time series anomaly detection. Finally, we conclude this work in Section VI.

II. METHODOLOGY

In this section, we provide the necessary information to help readers comprehend the motivation and highlight the design of *InoU*.

Data Pre-processing: In industrial setups, the deployed assets produce multivariate time series streams as they perform designated tasks. These multivariate data are usually sampled at a fixed or variable frequency to record metrics in various industrial processes. Due to the properties of these processes behind the streams, there may be significant differences in the numerical ranges of different channels.¹

The data stream is denoted as $\mathbf{x} = \{\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \mathbf{x}_{(3)}, \dots, \mathbf{x}_{(N)}\}$, $\mathbf{x} \in R^{N \times M}$, where N represents the total number of time steps, and M is the total number of features. Each one of its components $\mathbf{x}_{(t)}$ ($1 \leq t \leq N$) is a $1 \times M$ vector. A max-min scaler is then deployed to normalize $\mathbf{x}$, and use a sliding window with length $L = 128$ to segment the scaled time series data into slices. For example, we use $\mathbf{x}_{(t-L+1:t)} \in R^{L \times M}$ to denote a recording sequence that starts at time $t - L + 1$ and ends at time t .

Considering that a data segment is the minimal unit that the deep learning network consumes, we use $\mathbf{x}_t$ to represent $\mathbf{x}_{(t-L+1:t)}$ for a clear description in later parts of the paper. We also use $\mathbf{x}$ to represent the pre-processed dataset in instances where the dataset is mentioned, e.g., in the equations of the model optimization, but without the need to specify a specific timestep.

Anomaly Detection: Most prior anomaly detection works are built based on numerical error comparison. They rely on generative models to reconstruct data. Then, they calculate the deviation between the observed and the reconstructed values and determine if the sequences are anomalous based on the numerical segment-wise error. This criterion is based on the assumption that anomalies will have a more significant reconstruction error than normal data. However, there are a few things to note in actual practice:

¹By using the term “channel,” we refer to the univariate recordings in the multivariate time series data, with the intention to emphasize their physical meanings. This word may be used interchangeably with “feature” in the following parts.

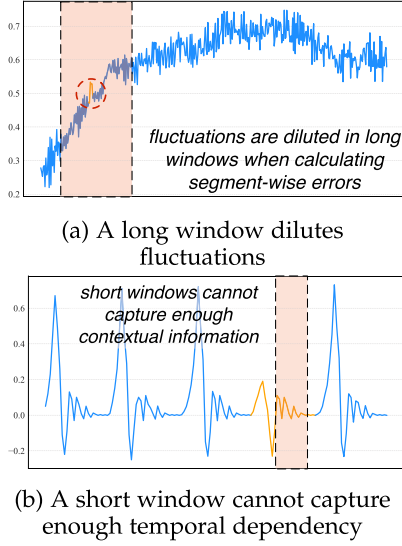


Fig. 2. Capture of temporal dependencies in segmented data. Orange sequences are anomalous.

1) *Segment length and limitations in dependency capture*: As stated, the operation of reconstruction and error calculation are all based on the length L that segments the original data stream. A longer length captures more temporal information but, on the other hand, dilutes fluctuations during the error calculation along the time axis (Fig. 2(a)). Meanwhile, a shorter length magnifies these fluctuations, but the limited temporal dependency that is captured may be insufficient to identify contextual or collective anomalies (Fig. 2(b)).

2) *Data normalization and deviation assessment*: Data normalization is an essential step in data preprocessing, ensuring that each channel is represented on a comparable scale. This technique helps neural networks to converge effectively, but it may also weaken the dynamics of each channel in the original data (Fig. 3(a)).

Under the principle of normalization, channels with high dynamic ranges will be processed with larger scaling factors. As such, fluctuations and irregularities in these channels become less salient after normalization, leading to potential neglect of anomalies (Fig. 3(b)). Besides, it can also be observed that the channels in these streams may have diverse physical meanings. This information is directly discarded since the error calculation treats all channels equally.

III. DESIGN OF INOU

A. Framework Overview

Fig. 4 shows the overall structure of *InoU*. This framework comprises two major parts: offline training and online detection.

Offline Training: During the training process, an efficient VAE (Variational Autoencoder) is trained on the unlabeled, noisy training materials. When this VAE model converges, it is again reused in the data pre-process pipeline. In this way, it serves as a filter that offsets the noisy and anomalous components in the training materials without the need for human intervention.

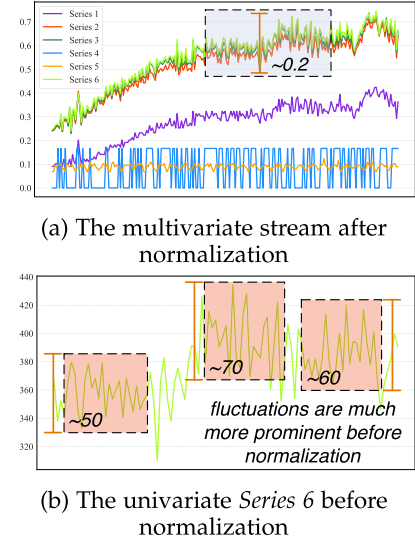


Fig. 3. Fluctuations in a high dynamic channel become less salient after normalization, leading to potential overlooks of anomalies.

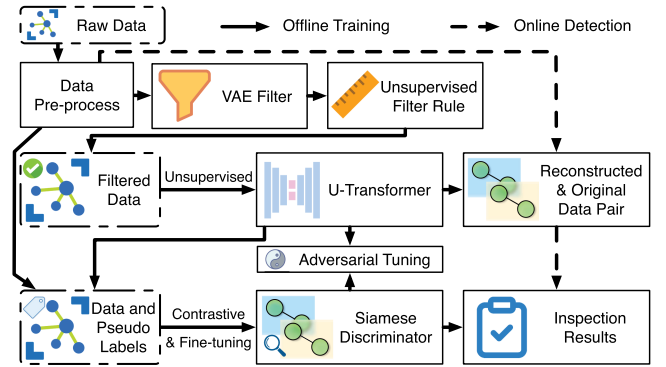


Fig. 4. Overall framework of *InoU*.

Based on the filtered training material, the U-Transformer is guided to extract the dependencies and reconstruct the data. During this step, the U-Transformer captures the essential representations of normal data, so their reconstructions can be done with high fidelity. However, the anomalies are less presented in the training materials, resulting in more significant reconstruction deviations. Driven by this principle, the U-Transformer is employed in two tasks: reconstructing the incoming sequences for further discrepancy analysis, and generating pseudo labels for the rest of the training process.

Utilizing contrastive learning techniques, the Siamese Discriminator is trained with two primary objectives: achieving high anomaly detection performance, and creating a manifold projection methodology that effectively metrics related to discrepancies even for samples that are unseen in historical recordings. The adversarial learning technique is further applied with a small learning rate, which optimizes embedding distributions by adjusting the manifold.

Online Detection: To identify anomalies in the data stream, the target samples are processed and reconstructed by the U-Transformer. Then, the Siamese Discriminator parallelly

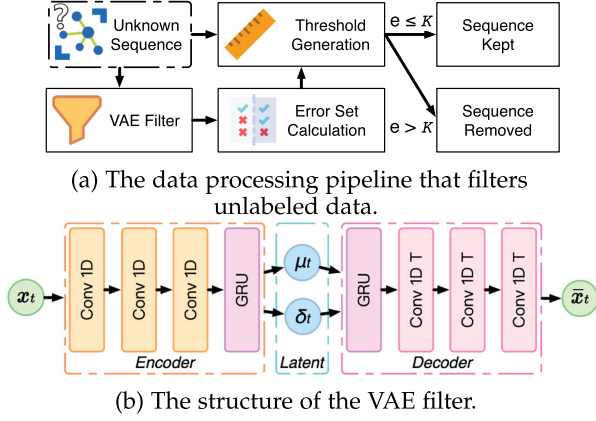


Fig. 5. The VAE model is used as a filter in the data processing pipeline.

examines the observed and reconstructed sample pairs, and compares their discrepancies at the embedding level. The final inspection results are thus generated.

B. VAE Driven Filtering of Unlabeled Training Material

Many prior works utilize VAE as the generative model to extract typical patterns from unlabeled training materials. However, some researchers [13], [14], [15], [16] have also pointed out that the VAE model based anomaly detection solutions may encounter performance issues when confronted with high noise levels, which is common in industrial setups. Besides, VAE models can also be biased in real-world events for several reasons [17]. This is due to the dilemma that the VAE model is trained to reconstruct the majority of training materials, but to what extent the data fitting should reach remains a challenging question.

To bypass this issue, we design a novel data processing pipeline (Fig. 5(a)) that decouples the VAE model from direct involvement in the anomaly detection task. The VAE model is employed to filter the training materials, rather than to reconstruct them. The underlying rationale of this design is intuitive. Since the unlabeled training materials exhibit significant imbalances, skewness, and noise in industrial scenarios, we deploy the VAE model as a filter to process the raw unlabeled training set $\mathcal{D}_{train,raw}$. This way, we form the filtered training set $\mathcal{D}_{train,flt}$ to ease impacts from the noisy and anomalous components. Details of this pipeline are as follows.

The VAE model (Fig. 5(b)) used for the processing pipeline mainly consists of two parts: the encoder and the decoder. The encoder in this VAE model interprets the input x_t into extracted data descriptions x_t^h , which are then used to generate the latent embeddings μ_t and δ_t .

$$x_t^h = \text{ReLU}(\text{Conv1D}(x_t, k3, s2)) \times 3 \quad (1)$$

$$\mu_t, \delta_t = \text{GRU}(x_t^h) \quad (2)$$

Here, we use *Conv1D* to denote the 1D convolution operation, whose arguments are *(input, kernel_size, stride)*, respectively. *ReLU* is the *Rectified Linear Unit* activation function that introduces non-linearity to the model. *GRU* is the *Gate*

Recurrent Unit layer, which bridges temporal dependencies. The operations of a *GRU* layer are presented as follows.

$$r_t = \sigma(W_r \cdot [\tilde{h}_{t-1} | x_t^h]) \quad (3)$$

$$u_t = \sigma(W_u \cdot [\tilde{h}_{t-1} | x_t^h]) \quad (4)$$

$$h_t = \tanh(W_h \cdot [r_t \odot \tilde{h}_{t-1} | x_t^h]) \quad (5)$$

$$\tilde{h}_t = (1 - u_t) \odot \tilde{h}_{t-1} + u_t \odot h_t \quad (6)$$

$$[\mu_t, \delta_t] = \tilde{h}_t \quad (7)$$

where r_t, u_t are the reset and update gate, $[\cdot]$ is the concatenation operation, and $\odot$ is the *Hadamard* product. Besides, $\sigma(\cdot)$ and $\tanh$ represent the *sigmoid* and *hyperbolic tangent* activation functions.

Based on the latent embeddings μ_t and δ_t , the decoder generates the reconstructed sample $\bar{x}_t$ by performing symmetrical operations as compared to the encoder.

$$x_t^z = \text{GRU}(\text{Rep}(\mu_t, \delta_t)) \quad (8)$$

$$\bar{x}_t = \text{ReLU}(\text{Conv1DT}(x_t^z, k3, s2)) \times 3 \quad (9)$$

Here, the term *Rep* denotes the *reparameterize* process. The arguments of *Conv1DT* (*transposed 1D convolution*) are similar to those of *Conv1D*, consisting of *(input, kernel_size, stride)*.

To optimize the parameters of this VAE filter model, we deploy a *Kullback-Leibler divergence* based loss:

$$\mathcal{L}_{flt} = \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] - \text{KL}[q_\phi(z|x) || p_\theta(z)] \quad (10)$$

Based on the trained VAE filter model, the error set $\mathbf{E}$ of the unlabeled training set $\mathcal{D}_{train,raw}$ is calculated as:

$$e_t = \frac{1}{L} \sum_{i=1}^L \|x_{(i)} - \bar{x}_{(i)}\|_2^2 \quad (11)$$

$$\mathbf{E} = \{e_t \mid 1 \leq t \leq N\} \quad (12)$$

where $\|\cdot\|_2$ is the *L2 Norm*, and L is the length of the window used to segment the data (Section II). $\bar{x}_{(i)}$ is the i -th time step inside the sequence being processed, in this case, the $\bar{x}_t$.

A threshold κ for the filtering process is then determined based on the distribution of $\mathbf{E}$. By applying the filter, a filtered training set $\mathcal{D}_{train,flt}$ is generated.

$$x_t \in \mathcal{D}_{train,flt}, \text{ if } e_t \leq \kappa \quad (13)$$

C. Trio-Attention U-Transformer for Representation Bridging and Data Reconstruction

Overview: Enlightened by advancements in U-Net [18], [19] and transformer [20], we design a trio-attention U-Transformer structure (Fig. 6) that is optimized for multivariate time series representation bridging and reconstruction while keeping outstanding scalability.

The U-Transformer is employed to construct the typical representations from the filtered training set $\mathcal{D}_{train,flt}$. Since the learned patterns are derived from the filtered data, the U-Transformer is expected to amplify discrepancies between the

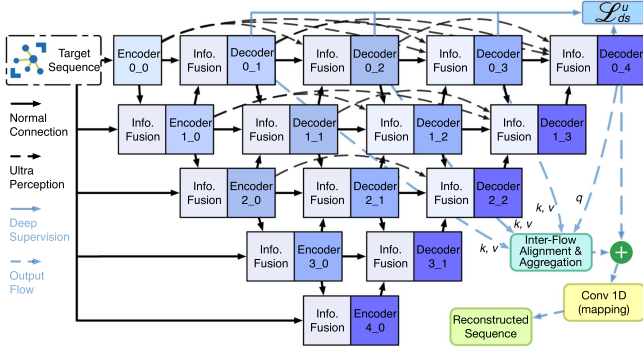


Fig. 6. The structure of the trio-attention U-Transformer.

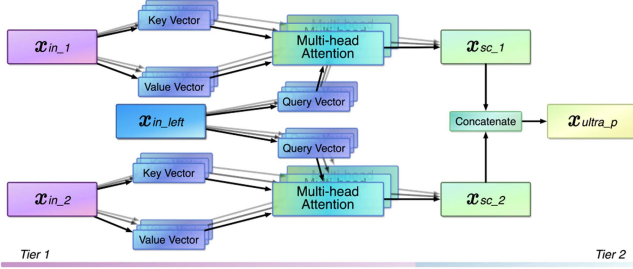


Fig. 7. The two-tier information flow aggregation in each block of the U-Transformer.

input and output when anomalies occur. To achieve this goal, we weave a delicate structure (Fig. 6) with multiple tricks for information extraction and aggregation.

Ultra Perception Shortcut Attention Driven Two-Tier Information Flow Aggregation: One key feature of the nested structure in U-Transformer is the long shortcut connections that allow information to flow between different levels of the network. To enhance the aggregation of information from shallower or deeper levels that is processed with different granularity, we design a two-tier information flow aggregation mechanism that is based on the ultra perception shortcut attention technique (Fig. 7).

For clarity in notation, we use x_{in_left} for the output of the horizontal left block of the current block where this attention mechanism is applied. Similarly, we use x_{in_o} to denote the output of blocks that are connected to the current block through shortcut connections, where o in the subscript indicates their sequence number. We denote the total number of these shortcut connected blocks as ω .

In detail, the ultra perception aggregation mechanism functions in a two-tier manner. In the first tier, the fusion of information flow performs in a block-to-block way. During this process, the remote information from the shortcut connected blocks is compared and aggregated with the information from the horizontal left block. Take the fusion between x_{in_left} and x_{in_1} as an example.

$$x_{sc_1} = \text{LN} \left[\rho_1^{(1)}, \rho_1^{(2)}, \dots, \rho_1^{(i)}, \dots, \rho_1^{(H)} \right] \quad (14)$$

Here, LN denotes the *layer normalization* operation that helps to stabilize the training process, $[\cdot]$ is the concatenation operation, and H is the total number of attention heads. $\rho_1^{(i)}$ is the

output of the i -th head in the multi-head attention mechanism, which is calculated as follows:

$$\rho_1^{(i)} = \text{Att} \left(W_q^{(i)} x_{in_left}, W_k^{(i)} x_{in_1}, W_v^{(i)} x_{in_1} \right) \quad (15)$$

where $W_q^{(i)}$, $W_k^{(i)}$, and $W_v^{(i)}$ are the query, key, and value weights for the i -th head, respectively. *Att* is the dot-product attention mechanism:

$$\text{Att}(q, k, v) = \text{Softmax}(qk^T) v \quad (16)$$

In the second tier, the information fusion focuses on aggregating information from different flows. This step helps the U-Transformer capture significant patterns at different granularities without losing a global perspective on the data stream.

$$x_{ultra_p} = [x_{in_left}, x_{sc_1}, x_{sc_2}, \dots, x_{sc_w}] \quad (17)$$

Intra-Flow Attention Driven Feature Map Contraction and Expansion: To guide the model to generate representations that are more informative, we use this technique to further process the output from the ultra perception shortcut attention mechanism.

The specific process of this operation varies depending on the location of the flow. Two main paths form the universal data flow in the U-Transformer: the contraction path and the expansion path. The former bridges complex dependencies by compressing the incoming sequences. Meanwhile, the latter weaves the information coming from various flows, and fuses them adaptively. In this way, the U-Transformer is able to keep a high level of scalability while extracting the most significant patterns from the data.

On the contraction path, the aggregated flow x_{ultra_p} is processed by two cascaded 1D convolutional (*Conv1D*) layers:

$$x_{if} = \text{Conv1D+S}(\text{Conv1D+S}(x_{ultra_p}, k3, s1), k3, s2) \quad (18)$$

Here, Conv1D+S denotes the *Conv1D* operation with *SeLU* (Scaled Exponential Linear Unit) activation. Then, a follow-up multi-head attention module is applied to the output x_{if} for fine-grained dependency extraction.

$$x_{is_fmc} = \text{LN} \left[\rho_{ic}^{(1)}, \rho_{ic}^{(2)}, \dots, \rho_{ic}^{(i)}, \dots, \rho_{ic}^{(H)} \right] \quad (19)$$

where H is the total number of attention heads. $\rho_{ic}^{(i)}$ is the output of the i -th head in the multi-head attention mechanism to process the information in x_{if} (18):

$$\rho_{ic}^{(i)} = \text{Att} \left(W_q^{(i)} x_{if}, W_k^{(i)} x_{if}, W_v^{(i)} x_{if} \right) \quad (20)$$

On the expansion path, the resolution of the feature map gradually gets restored through the cascaded transposed 1D convolutional (*Conv1DT*) layers.

$$x_{if} = \text{Conv1DT+S}(\text{Conv1DT+S}(x_{ultra_p}, k3, s2), k3, s1) \quad (21)$$

Here, Conv1DT+S denotes the *Conv1DT* operation with *SeLU* activation. This operation is also followed by a multi-head attention module:

$$\rho_{ie}^{(i)} = \text{Att} \left(W_q^{(i)} x_{if}, W_k^{(i)} x_{if}, W_v^{(i)} x_{if} \right)$$

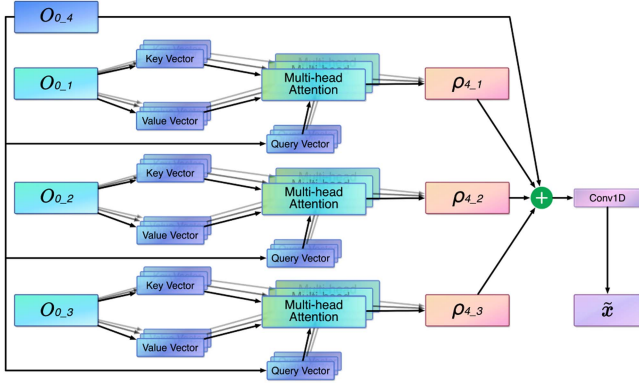


Fig. 8. The alignment and aggregation of output flows in the U-Transformer.

$$\mathbf{x}_{is_fme} = \text{LN} \left[\rho_{ie}^{(1)}, \rho_{ie}^{(2)}, \dots, \rho_{ie}^{(i)}, \dots, \rho_{ie}^{(H)} \right] \quad (22)$$

Inter-Flow Attention Driven Output Alignment and Aggregation: Owing to the unique architecture, the U-Transformer model demonstrates the capability to extract patterns at various levels of granularity while maintaining excellent scalability. However, in actual practice, the effective regulation and alignment of the output nodes are crucial for ensuring the model's performance. To address this challenge, we propose an inter-flow attention driven output alignment and aggregation mechanism (Fig. 8).

For clear descriptions, we denote the output nodes, i.e., the nodes on the top level of the U-Transformer that generate reconstruction results, as $O_{0,1}$ to $O_{0,4}$. Take the aggregation between $O_{0,4}$ and $O_{0,1}$ as an example, the output alignment happens as follows:

$$\rho_{4,1} = \text{LN} \left[\rho_{4,1}^{(1)}, \rho_{4,1}^{(2)}, \dots, \rho_{4,1}^{(i)}, \dots, \rho_{4,1}^{(H)} \right] \quad (23)$$

where the calculation of the output from different heads is performed as follows:

$$\rho_{4,1}^{(i)} = \text{Att} \left(\mathbf{W}_q^{(i)} O_{0,4}, \mathbf{W}_k^{(i)} O_{0,1}, \mathbf{W}_v^{(i)} O_{0,1} \right) \quad (24)$$

A more general aggregation that forms the fused output is then conducted:

$$\rho_{0aa} = O_{0,4} \oplus \rho_{4,3} \oplus \rho_{4,2} \oplus \rho_{4,1} \quad (25)$$

Here, $\oplus$ denotes the in-place addition operation. Finally, the reconstruction result $\tilde{x}$ is obtained by applying a mapping operation:

$$\tilde{x} = \text{Conv1D+S}(\rho_{0aa}, k1, s1) \quad (26)$$

Deep Supervision Guided Model Training: Since its introduction, deep supervision has been widely explored for improving the performance of various downstream tasks [21], [22], [23]. We adapt this technique to the U-Transformer structure to regulate the convergence of the intermediate layers. Take the output from the output node $O_{0,4}$ for example, we impose a similarity loss to guide its output:

$$\mathcal{L}_{0-4}^u = \sum_{t=1}^N \|\mathbf{x}_t - O_{0,4}\|_2^2 \quad (27)$$



Fig. 9. The structure of the Siamese discriminator.

Here, $\|\cdot\|_2$ is the $L2$ Norm. To help the intermediate layers of the U-Transformer converge effectively and perform in an aligned way, we further impose a deep supervision loss to the output node set:

$$\begin{aligned} \mathcal{L}_{ds}^u = & \eta_0 \mathcal{L}_{0-a}^u + \eta_1 \mathcal{L}_{0-1}^u + \eta_2 \mathcal{L}_{0-2}^u \\ & + \eta_3 \mathcal{L}_{0-3}^u + \eta_4 \mathcal{L}_{0-4}^u \end{aligned} \quad (28)$$

where $\eta_0 - \eta_4$ are the seasoning weights. $\mathcal{L}_{0-a}^u$ denotes the similarity loss applied to the aggregated output in (26).

D. Siamese Discriminator for Embedding Level Discrepancy Analysis

Overview: As discussed in Section II, most prior works use segment-wise numerical reconstruction error to identify anomalies. This scheme is less effective for industrial data, especially the one featuring high dimensionality and high levels of noise and anomalies. To tackle this unique challenge in industrial scenarios, we design a Siamese discriminator (Fig. 9) that projects the observed and reconstructed sequences into manifold space, and adaptively collates their discrepancies at the embedding level. As such, *InoU* gains the ability to inspect even the most subtle inconsistencies between the observed and reconstructed sequences, which is crucial for improving anomaly detection performance in industrial settings.

Manifold Projection: To project the target sequences into the abstract manifold, we design a Siamese backbone that processes the observed and the reconstructed sequences with a shared set of parameters in parallel. We define an elementary block in the Siamese backbone as follows:

$$\begin{aligned} \mathbf{x}_{se1} &= \text{Conv1D+S}(\mathbf{x}_{in}, k3, s1) \\ \mathbf{x}_{se2} &= \text{Res1D}(\mathbf{x}_{se1}) \\ \mathbf{x}_{se3} &= \text{BN}(\text{MP}(\mathbf{x}_{se2}, 2)) \end{aligned} \quad (29)$$

Here, $\mathbf{x}_{in}$ is the input of this elementary block, BN is the *batch normalization* operation, and MP is the *max pooling* operation. Res1D denotes the *1D residual* module, which operates as:

$$\mathbf{x}_{se2} = \mathbf{x}_{se1} \oplus \text{Conv1D+S}(\mathbf{x}_{se1}, k3, s1) \times 3 \quad (30)$$

Based on the elementary block (“Elem” in later parts) described in (29)–(30), we further form the backbone of the Siamese discriminator. Since the primary function of this backbone is to extract and expose the most salient features for subsequent discrepancy analyses, its architecture needs to be carefully crafted to balance feature extraction efficiency and the ability to capture both low-level and high-level embeddings. To satisfy this need, we design two distinct output structures for different purposes: the representation output $\mathcal{R}$ that generates

the projected manifold of the incoming sequence, and the projection head $\mathcal{P}$ that has a relatively deeper structure, and is used to regulate the convergence.

The representation output is generated as:

$$\mathcal{R} = \text{Elem}(\mathbf{x}_{in}) \times 4 \quad (31)$$

Whereas the projection head is further formed through:

$$\begin{aligned} \mathbf{x}_{sp1} &= \text{Elem}(\mathcal{R}) \\ \mathbf{x}_{sp2} &= [\text{GRU}_f(\mathbf{x}_{sp1}), \text{GRU}_b(\mathbf{x}_{sp1})] \\ \mathcal{P} &= \text{BN}(\text{Conv1D} + \text{S}(\mathbf{x}_{sp2}, k3, s1)) \end{aligned} \quad (32)$$

Here, GRU_f and GRU_b represent the forward and backward calculations in a *Bidirectional GRU* module.

Inference on Anomalies: The employment of this Siamese architecture enables us to implement the downstream task of discrepancy comparison in an efficient yet robust way. By applying a shared set of parameters, i.e., the Siamese backbone, the target sequences are thus projected into an abstract manifold space that reveals pivotal metric information. Based on the projected manifolds, numerical inconsistencies of the concerned sequences that used to be subtle or less noticeable become much more pronounced during the embeddings level discrepancy comparison. In this way, the overall performance of anomaly detection is enhanced, particularly in industrial scenarios.

To describe this process, we notate a random normal sequence as $\mathbf{x}_n$, and a random anomalous one as $\mathbf{x}_a$. As the convergence of the U-Transformer happens on the filtered training material $\mathcal{D}_{train,flt}$, the reconstruction deviation of $\mathbf{x}_a$ will be significantly larger. Then we can have:

$$\text{Sim}(\mathcal{R}(\mathbf{x}_a), \mathcal{R}(\tilde{\mathbf{x}}_a)) \rightarrow 0 \quad (33)$$

where Sim is the calculation of cousin similarity:

$$\text{Sim}(\alpha, \beta) = \frac{\alpha^T \beta}{\tau_1 \|\alpha\| \|\beta\|} \quad (34)$$

in which $\|\cdot\|$ is the *L1 Norm*, and τ_1 is a coefficient to adjust this calculation. On the other hand, representations for a set of normal sequences result in a higher similarity:

$$\text{Sim}(\mathcal{R}(\mathbf{x}_n), \mathcal{R}(\tilde{\mathbf{x}}_n)) \rightarrow 1 \quad (35)$$

Based on the properties of $\mathcal{R}$, we cascade a discriminator to implement the anomaly detection task. Since the Siamese backbone processes both the observed sequence and the reconstructed one with a shared set of parameters, the discrepancies between the two sequences are thus reflected in the embeddings generated by the $\mathcal{R}$. The output of the backbone is concatenated as $\mathcal{R}_t^{\text{cct}}$:

$$\mathcal{R}_t^{\text{cct}} = [\mathcal{R}(\mathbf{x}_t), \mathcal{R}(\tilde{\mathbf{x}}_t)] \quad (36)$$

Then, several blocks are employed for metric analysis:

$$\begin{aligned} \mathbf{x}_{ma1} &= \text{BN}(\text{Conv1D} + \text{S}(\mathcal{R}_t^{\text{cct}}, k3, s2)) \\ \mathbf{x}_{ma2} &= \text{BN}(\text{Conv1D} + \text{S}(\mathbf{x}_{ma1}, k3, s1)) \\ \mathbf{x}_{ra} &= \text{Conv1D}(\text{Flatten}(\mathbf{x}_{ma2}), k1, s1) \end{aligned} \quad (37)$$

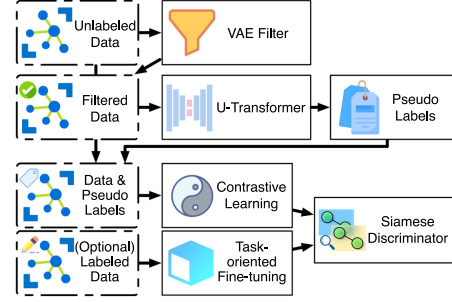


Fig. 10. Training workflow of the Siamese discriminator.

The inference results are generated through:

$$l_t = \text{Softmax}(\mathbf{x}_{ra}) \quad (38)$$

where l_t is the inference result for the input pair $\mathbf{x}_t$ and $\tilde{\mathbf{x}}_t$.

Training Workflow: To form an informative manifold space and ensure the full expression of metrics related to discrepancies, we design a contrastive learning based training workflow for the Siamese discriminator. As depicted in Fig. 10, this workflow comprises two stages: self-supervised contrastive learning and task-oriented fine-tuning.

1) *Self-supervised contrastive learning:* In this part, the Siamese discriminator is trained on unlabeled data in a self-supervised manner. The U-Transformer is first trained on the filtered training set $\mathcal{D}_{train,flt}$. Then, we use the U-Transformer to process the raw training set $\mathcal{D}_{train,raw}$, and calculate its error set $\mathcal{E}$. The error set is then used to calculate the threshold κ as stated in Section III-B, and thereafter generate the pseudo label set $\mathcal{J}_{psd}$. With $\mathcal{D}_{train,raw}$ and $\mathcal{J}_{psd}$, we use the output from the projection head in the Siamese discriminator to form the loss for matched pairs:

$$\begin{aligned} \text{Dis} &= 1 - \text{Sim}(\mathcal{P}(\mathbf{x}_t), \mathcal{P}(\tilde{\mathbf{x}}_t)) \\ \mathcal{L}_{match} &= \text{Dis} \times (1 - y_{t,psd}) \end{aligned} \quad (39)$$

where $y_{t,psd}$ is the pseudo label of $\mathbf{x}_t$.

Similarly, for the mismatched pairs, we have:

$$\mathcal{L}_{mismatch} = \text{Sim}(\mathcal{P}(\mathbf{x}_t), \mathcal{P}(\tilde{\mathbf{x}}_t)) \times y_{t,psd} \quad (40)$$

By integrating the two components as in (39)–(40), we form the contrastive loss for the Siamese backbone:

$$\mathcal{L}_{sum} = \tau_2 \mathcal{L}_{match} + \tau_3 \mathcal{L}_{mismatch} \quad (41)$$

where τ_2 and τ_3 are the seasoning weights.

2) *Task-oriented fine-tuning:* Apart from the contrastive regulation being applied to the Siamese backbone, we utilize other techniques to assist in the formation of the manifold. Considering that in some industrial scenarios, a limited amount of labeled data may be available, a binary cross entropy loss is applied:

$$\begin{aligned} \mathcal{L}_{bce}^d &= -\frac{1}{N} \sum_{t=1}^N [y_t \cdot \log(p(y_t)) \\ &\quad + (1 - y_t) \cdot \log(1 - p(y_t))] \end{aligned} \quad (42)$$

Here, y_t is the ground-truth label of the limited dataset. Besides, a center loss [24] is also employed to enhance the inter-class sparsity:

$$\mathcal{L}_{ctr}^d = \frac{1}{N} \sum_{t=1}^N \|\mathcal{R}(\mathbf{x}_t) - \mathbf{c}_{y_t}\|_2^2 \quad (43)$$

where $\mathbf{c}_{y_t}$ is the cluster for $\mathbf{x}_t$ in the feature space. Based on (42)–(43), the following sum loss is formed to optimize the Siamese discriminator:

$$\mathcal{L}_{sum}^d = \mathcal{L}_{bce}^d + \mathcal{L}_{ctr}^d \quad (44)$$

However, scenarios with no available human labeled resources are also common in industrial applications. To address this situation, the ground-truth labels y_t can be replaced with the pseudo labels $y_{t,psd}$. This workaround also effectively supports the training of *InoU*, as it is validated later in Section IV-D.

E. Manifold Adjusting Via Adversarial Learning

In industrial scenarios, on-site data usually contain a considerable proportion of noise and anomaly components. Besides, historical recordings may not cover all cases in actual deployment. To enhance *InoU*'s ability to spot historically unobserved samples and to improve the Siamese discriminator's capacity for metric expression in the manifold space, we introduce an adversarial learning based manifold adjusting scheme.

With the two primary components, i.e., U-Transformer and the Siamese discriminator, being integrated under the unified framework, *InoU* naturally fits the concept of an adversarial architecture. Driven by this finding, the following adversarial loss is applied:

$$\begin{aligned} \mathcal{L}_{sum}^{adv} = & \min_{\mathcal{M}_{U-Tran}} \mathcal{L}_{ds}^u + \min_{\mathcal{M}_{Siamese}} \mathcal{L}_{sum}^d \\ & + \max_{\mathcal{M}_{U-Tran}} \min_{\mathcal{M}_{Siamese}} \tau_2 \mathcal{L}_{match} \\ & + \min_{\mathcal{M}_{Siamese}} \tau_3 \mathcal{L}_{mismatch} \end{aligned} \quad (45)$$

To ensure that the minor yet stable alternation to the learned parameters, the adversarial loss is applied with a small learning rate. The effects of this adversarial pipeline are validated and illustrated in Section IV-H.

F. Anomaly Cause Localization

In the middle of an undesirable event, localization of anomaly cause assists engineers in identifying the most probable origin of the incident. As such, the process to re-gain control of the case can be expedited. To realize this goal, we analyze the intermediate vector $\mathbf{x}_{ra}$ in the Siamese discriminator (37), and infer their contribution to the formation of an anomalous event. The visualization in Fig. 11 is presented on an anomalous sample.

IV. EVALUATION

A. Datasets and Metrics

Dataset: We incorporate five large-scale industrial datasets to evaluate the performance of *InoU* under different industrial

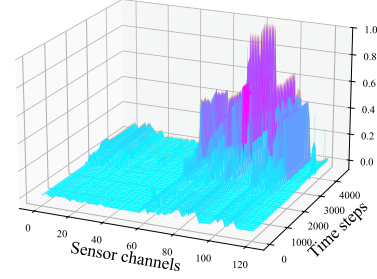


Fig. 11. Anomaly cause visualization.

TABLE I
DATASET PROPERTIES

Dataset	SMD	SWaT	WADI	SMAP	MSL
Monitored Assets	28	/	/	55	27
Number of Features	38	51	123	25	55
Observation Duration	5 Weeks	11 Days	16 Days	12 Months	/
Anomaly Portion	4.16 %	12.14 %	5.77 %	13.13 %	10.53 %
Scenarios	Cloud Center	Water Treatment	Water Treatment	Satellite Telemetry	Satellite Telemetry

scenarios. These datasets are widely used in many other anomaly detection works.

1) *SMD* [10]: This dataset contains KPI (Key Performance Indicator) recordings from a computing center of a cloud service provider.

2-3) *SWaT* [25] & *WADI* [25]: The Secure Water Treatment (SWaT) and the Water Distribution (WADI) are two datasets that record the operation status of attacked water treatment systems.

4-5) *SMAP* [26] & *MSL* [26]: The Soil Moisture Active Passive (SMAP) and the Curiosity Rover on Mars (MSL) datasets contain telemetry recordings from spacecrafts.

The properties of these datasets are listed in Table I.

Metrics: We use the metric set as in other works to reflect a concerned system's overall anomaly detection performance: *Precision*, *Recall*, and *F1-Score*. Their calculations are as follows: $\text{Precision} = \frac{TP}{TP+FP}$, $\text{Recall} = \frac{TP}{TP+FN}$, and $\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$.

B. Baselines and the Test Environment

Baselines: We compare the performance of *InoU* with state-of-the-art works: *VAE-LSTM* [9], *OmniAnomaly* [10], *USAD* [11], *InterFusion* [12], *CAE-M* [13], *AMSL* [15], *TranAD* [27], and *AnoTran* [28]. We also employ a classical algorithm, *Isolation Forest*, for performance comparison, though not involve it in the discussions.

Environment: We use a workstation to conduct the evaluation. Its main configurations are listed in Table II.

Parameters: Table III lists the parameters used to train *InoU*. The arguments are left as the default if not otherwise specified.

TABLE II
TESTBED CONFIGURATION

Config	Detail
CPU	AMD EPYC 7742
RAM	192 GB ECC DDR4
Storage	64 TB U.2 NVMe
GPU	NVIDIA RTX A6000

TABLE III
PARAMETERS

Parameters	Settings
Batch Size	32
Window Length	128
Epochs	500 (with early stopping)
Optimizer	Adamax (lr=3e-3)

C. Performance and Analysis

In this section, we compare the anomaly detection performance of *InoU* with the baselines. During this experiment, all results are the average of three randomized trials. The overall performance results are presented in Table IV.

It can be observed that *InoU* outperforms the baselines on every dataset in terms of *F1-Score*. On these large, real-world, industrial datasets, *InoU* gets the best average *F1-Score* of 0.9168, with an increment of 5.58% compared to the state-of-the-art. Among these results, several findings should be noted. All baselines' performance slightly drops on the *WADI* dataset, possibly due to its low anomaly portion and high dimension. *InoU*, however, achieves a boost in *F1-Score* of 28.20% on this dataset. This finding echoes the observations on the *SWaT* and *MSL* datasets, which are characterized by their high dimensionality. The utilization of the Siamese discriminator in *InoU* also yields significant improvements in *F1-Score* on the two datasets: the *SWaT* dataset with 51 features and the *MSL* dataset with 55 features. Respectively, *InoU* achieves a performance improvement in *F1-Score* by 3.08% and 2.53% on these two datasets.

Moreover, our proposed work *InoU* demonstrates significant performance improvement, even on datasets where the state-of-the-art models already perform well. On the *SMD* and *SMAP* datasets, *InoU* achieves a substantial increase in the *F1-Score* by 1.08% and 1.64%, respectively. These results validate the effectiveness of *InoU* across various datasets, highlighting its robust performance in industrial anomaly detection.

Apart from the results suggested by the *F1-Score*, we can also observe the robustness of *InoU* from other aspects. It is clear from Table IV, *InoU* achieves the best *recall* performance on *SMD*, *SWaT*, *SMAP*, and *MSL*, and its *recall* is the second-best on *WADI*. A higher *recall* is crucial in industrial applications, as it represents the ability to accurately detect all anomalous samples without missing any. The improvement in *recall* indicates a notable reduction in the number of undetected anomaly samples

compared to these baselines, enabling engineers to identify and address potential issues promptly.

We continue discussing the effectiveness of our design in *InoU* in the following sections.

D. Validation of *InoU* When Strictly Unsupervised

InoU is designed to utilize unlabeled industrial datasets to the greatest extent. By developing a training pipeline that incorporates a VAE filter, contrastive / adversarial learning, and task-oriented fine-tuning, the robustness of *InoU* gets improved. In this section, we use ablation studies to discuss the potential impacts caused by the materials used in the fine-tuning process of the Siamese discriminator. Three cases are introduced in this experiment:

- 1) Best Baseline: the results of the best baseline, namely *AnoTran* [28].
- 2) Strictly Unsupervised: the calculation of (42) relies only on the pseudo label set $\mathcal{J}_{psd}$ that the U-Transformer produces. In this case, no human labeled resource is involved.
- 3) Fine-tuned: the original training workflow as introduced in Section III-D.

In this ablation study, we select three industrial datasets covering low, medium, and high numbers of features to obtain a more generalizable conclusion. The experiment results are presented in Table V.

As suggested by the data, a slight performance drop can be observed when *InoU* is trained on a fully unlabeled dataset. It is clear that the effectiveness of the strict unsupervised manner shows connections with characteristics of the dataset being processed. In the case of strictly unsupervised manner, high-dimensional dataset, e.g., *WADI*, may experience a more brutal performance impact than those with fewer features, such as *SWaT* and *SMAP*. This finding indicates that expert domain knowledge and human-labeled materials continue to play a significant role in enhancing the performance of anomaly detection systems.

However, it is worth noting that *InoU*, when trained strictly unsupervised, still beats the best baseline on these datasets, with an increment in the average *F1-Score* of 5.40%. This is to say, though removing limited labeled data from the training materials when fine-tuning the Siamese discriminator may lead to a slight performance impact, *InoU* still provides a competitive and robust performance for the anomaly detection task in industrial environments.

E. Ablation Study on Critical Components

Under the framework of *InoU*, many critical components are designed to work seamlessly and function effectively. In this section, we design an ablation study to assess their impacts on the overall performance of *InoU*. The concerned components include:

- 1) deep supervision in the U-Transformer (28);
- 2) implementation of the recurrent layers in Siamese discriminator (32);
- 3) the filter in the data pre-processing workflow (Section III-B);

TABLE IV
ANOMALY DETECTION PERFORMANCE. MP: MEAN *PRECISION*, MR: MEAN *RECALL*, mF1: MEAN *F1-SCORE*

Approaches	SMD (IT Company)			SWaT (Water Equipment)			WADI (Water Equipment)		
	mP	mR	mF1	mP	mR	mF1	mP	mR	mF1
<i>IF</i>	0.5909	0.8302	0.6904	0.7476	0.6444	0.6922	0.2618	0.3983	0.3160
<i>VAE-LSTM</i> [9]	0.8181	0.8962	0.8554	0.8909	0.6035	0.7196	0.2119	0.8942	0.3426
<i>OmniAnomaly</i> [10]	0.9340	0.9528	0.9433	0.9805	0.6558	0.7859	0.4283	0.4119	0.4200
<i>USAD</i> [11]	0.9274	0.9709	0.9486	0.9805	0.6953	0.8137	0.4333	0.4206	0.4269
<i>InterFusion</i> [12]	0.9438	<u>0.9825</u>	0.9627	0.9869	0.6957	0.8161	0.3931	0.5000	0.4401
<i>CAE-M</i> [13]	0.9293	0.9435	0.9364	0.9762	0.6520	0.7818	0.4117	0.4301	0.4207
<i>AMSL</i> [15]	0.9429	0.9533	0.9481	<u>0.9852</u>	0.6956	0.8154	0.4331	0.4543	0.4434
<i>TranAD</i> [27]	0.9431	0.9817	0.9620	0.9831	0.7102	0.8246	0.4333	0.4621	0.4473
<i>AnoTran</i> [28]	<u>0.9465</u>	0.9800	<u>0.9630</u>	0.9698	<u>0.8620</u>	<u>0.9128</u>	<u>0.5339</u>	0.6188	<u>0.5732</u>
<i>InoU</i>	0.9505	0.9974	0.9734	0.9776	0.9068	0.9409	0.6773	<u>0.8032</u>	0.7349

Approaches	SMAP (Solar Telemetry)			MSL (Solar Telemetry)			Average			
	mP	mR	mF1	mP	mR	mF1	mP	mR	mF1	Δ
<i>IF</i>	0.4622	0.5354	0.4961	0.6083	0.6540	0.6303	0.5342	0.6125	0.5650	61.63%
<i>VAE-LSTM</i> [9]	0.8318	0.8610	0.8462	0.7654	0.7699	0.7676	0.7036	0.8050	0.7063	77.03%
<i>OmniAnomaly</i> [10]	0.7893	0.9312	0.8544	0.8696	<u>0.9576</u>	0.9115	0.8003	0.7819	0.7830	85.40%
<i>USAD</i> [11]	0.8466	0.8881	0.8668	<u>0.9300</u>	0.9223	0.9261	0.8236	0.7794	0.7964	86.87%
<i>InterFusion</i> [12]	0.8551	0.9247	0.8885	0.8977	0.9557	0.9258	0.8153	0.8117	0.8067	87.98%
<i>CAE-M</i> [13]	0.8940	0.8815	0.8877	0.9142	0.9167	0.9155	0.8251	0.7648	0.7884	85.99%
<i>AMSL</i> [15]	0.8914	0.8978	0.8946	0.9033	0.9291	0.9160	0.8312	0.7860	0.8035	87.64%
<i>TranAD</i> [27]	0.9007	0.9113	0.9060	0.9119	<u>0.9576</u>	<u>0.9342</u>	0.8344	0.8046	0.8148	88.87%
<i>AnoTran</i> [28]	<u>0.9446</u>	<u>0.9788</u>	<u>0.9614</u>	0.9102	0.9538	0.9315	<u>0.8610</u>	<u>0.8787</u>	<u>0.8684</u>	94.71%
<i>InoU</i>	0.9623	0.9925	0.9772	0.9392	0.9773	0.9579	0.9014	0.9354	0.9168	100.00%

The best results are in bold, and the second-best ones are underlined.

TABLE V
PERFORMANCE COMPARISON IN A STRICTLY UNSUPERVISED MANNER

F1-Score	SWaT	Comp	WADI	Comp.
Best Baseline	0.9128	97.01 %	0.5732	78.00 %
Strictly Unsupervised	0.9225	98.04 %	0.6899	93.87 %
Fine-tuned	0.9409	100.00 %	0.7349	100.00 %

F1-Score	SMAP	Comp.	Average	Avg. Comp.
Best Baseline	0.9614	98.38 %	0.8158	92.25 %
Strictly Unsupervised	0.9673	98.99 %	0.8599	97.24 %
Fine-tuned	0.9772	100.00 %	0.8843	100.00 %

To finish these experiments, the following configurations are deployed correspondingly:

C1) we deactivate deep supervision on nodes $O_{0,1}$ to $O_{0,3}$ (C1-NS), keep deep supervision on nodes $O_{0,3}$ and $O_{0,4}$ (C1-HS), and compare these two variants with the fully activated deep supervision in *InoU* (C1-DS).

C2) we compare two other variants of recurrent structure, i.e., *RNN* (Recurrent Neural Network) and *LSTM* (Long Short-Term Memory) with the *GRU* module used in our work. These variants are marked as C2-RNN, C2-LSTM, and C2-UM, respectively.

C3) we use the *Spectral Residual* technique to replace the *VAE* filter in our data processing pipeline. Results are noted as

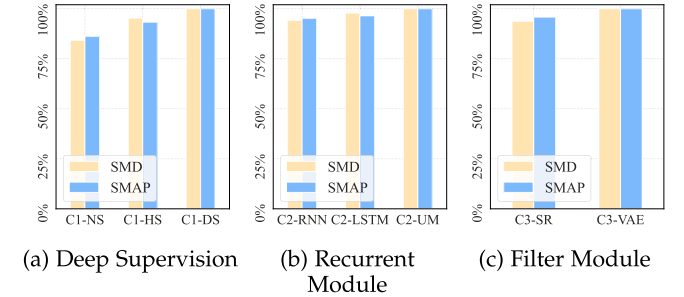


Fig. 12. Ablation study on the effects of critical components.

C3-SR (Spectral Residual), and C3-VAE (our original version that enables the *VAE* filter).

We provide the results in Fig. 12. Listed below are the key findings:

Effects of the deep supervision: The removal of deep supervision results in noticeable differences in performance on the tested datasets. For the C1-HS, the drop of *F1-Score* is less significant on *SMD* (4.74%) than on *SMAP* (6.76%). On the other hand, the C1-NS variant performs better on *SMAP* with a drop of 13.94%, milder than the one on *SMD* (15.82%). As the results reflect, it is straightforward that deep supervision helps

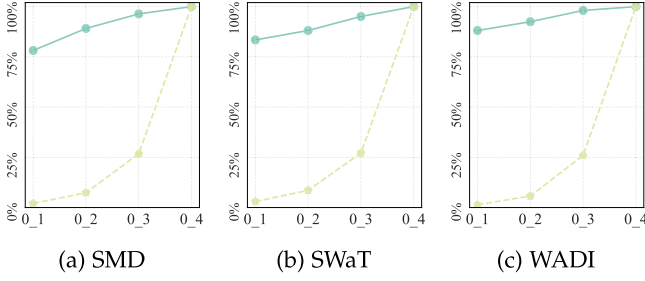


Fig. 13. Performance and computation overhead of different variants of the U-Transformer. Solid line: *F1-Score*; Dashed line: Parameter amount.

regulate the update of parameters in the intermediate layers and gains a better performance.

Effects of the recurrent module: The results across the datasets indicate that *GRU* in our model is more efficient than its two variants in bridging temporal connections. This finding is intuitive for the following reasons. As discussed in [29], the *RNNs* suffer from gradient problems, potentially limiting their ability to capture long-term dependencies. Due to the trio-gate design, the *LSTM* network is computationally heavy, which may lead to difficulties converging on noisy materials. These factors may contribute to the performance limitations of these two variants.

Effects of the filter module: Industrial data usually contain noise and anomalous samples, whereas human labeling is laborious and time-consuming. To elevate anomaly detection performance in such scenarios, we introduce an efficient *VAE* filter in the data pre-processing pipeline to offset the impacts of noised and anomalous samples. Though *Spectral Residual* offers a solution for data cleaning, its insufficient utilization of contextual information can result in a performance drop when processing complex industrial data.

F. Scalability of the U-Transformer

The U-Transformer in the *InoU* framework is designed with a nested structure, enabling this structure to capture multi-scale information while maintaining excellent scalability and efficiency. To evaluate the scalability of the U-Transformer, we conduct experiments to assess its performance under different trimming settings. To be specific, we capture the output from O_{0_1} to O_{0_4} , and connect these outputs to the Siamese discriminator for the task of anomaly detection. The performance of trimmed versions of the U-Transformer is compared with that of the full-size model. We present the corresponding results in Fig. 13.

It is straightforward that, the original, full-sized model that outputs from O_{0_4} achieves the best anomaly detection results on the tested dataset. However, this output flow also incurs the heaviest computation overhead, as it activates most of the parameters in the U-Transformer. On the contrary, O_{0_1} is the most computation-efficient flow in the nested structure. From these figures, we can observe a parameter reduction of up to 97% with a general 10% - 20% performance drop. For the two intermediate output nodes O_{0_2} and O_{0_3} , a more flexible solution can be achieved between computation cost and performance, addressing the needs for varied deployment targets. The

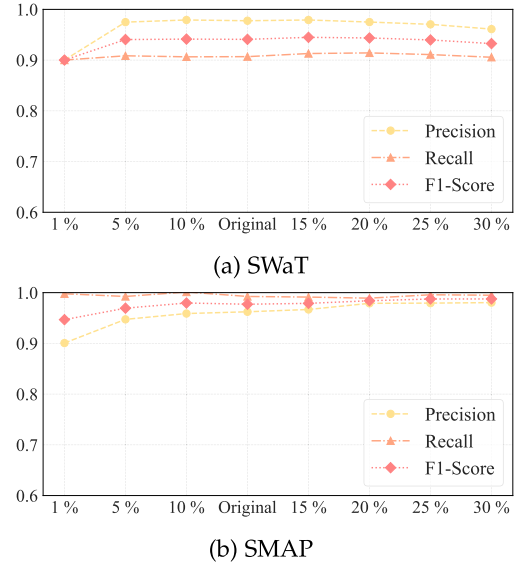


Fig. 14. Performance across different anomaly portions.

O_{0_3} provides a more efficient solution (25% parameter amount with 98% performance). As for O_{0_2} , its parameter amount is comparable to O_{0_1} , but comes up with a 4% to 14% increment in *F1-Score*.

To summarize, the nested structure built in U-Transformer enables the model to sense and capture multi-scale context information from different levels. Thus, it is possible to adjust the overall computation overhead by altering the flow through which data are passed. By doing so, the model can be adapted to various deployment scenarios, while maintaining a high level of anomaly detection performance.

G. Impacts of Anomaly Portion

The datasets used for experiments exhibit a noticeable imbalance, with anomalies being substantially underrepresented compared to normal samples. This disparity reflects the real-world scenario where anomalies are typically rare events. However, it is important to note that the frequency of anomalies can fluctuate considerably in industrial environments due to the diverse and often unpredictable operational conditions of on-site equipment and processes.

To assess the robustness of *InoU* across varying levels of anomaly percentages, we conduct experiments on two representative datasets, namely the *SWaT* and *SMAP*. The test set is varied to include different proportions of anomalies, ranging from 1% to 30%. Specifically, we evaluate performance at seven distinct anomaly levels: 1%, 5%, 10%, 15%, 20%, 25%, and 30% anomalies. *Precision*, *Recall*, and *F1-Score* results of this experiment are presented in Fig. 14.

In general, *InoU* maintains a high availability of anomaly detection across different anomaly portions. At the extremely low anomaly percentage (1%), a decrease can be observed in the performance of *InoU* on both datasets. On the *SWaT* dataset, the *F1-Score* drops to 0.9001 when the anomaly percentage is set to 1%. The decrease in *SMAP* is less pronounced, with the

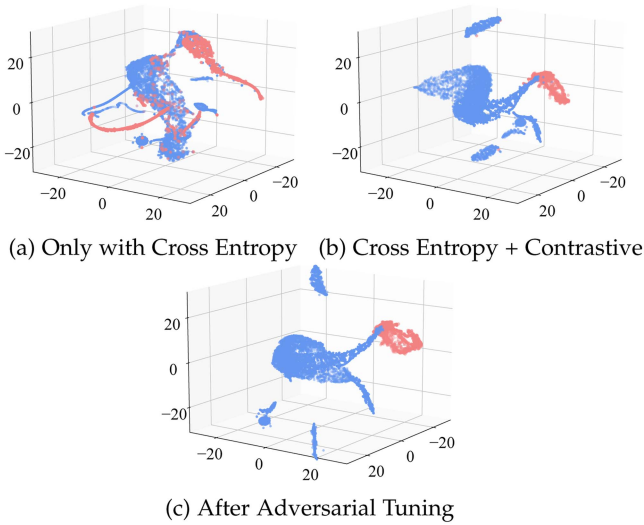


Fig. 15. Visualization of the Manifold Space of the Projection Head in the Siamese Discriminator. Blue points: representations of normal samples; Red points: representations of anomalies.

F1-Score keeping at a relatively high level of 0.9467. However, in other cases, the anomaly detection performance of *InoU* remains stable. For anomaly percentages from 5% to 30%, the *F1-Score* of *InoU* on the two datasets is consistently high, fluctuating around 0.94 and 0.98 on *SWaT* and *SMAP*, respectively. This finding suggests that *InoU* is capable of maintaining a robust anomaly detection performance across a wide range of anomaly percentages. Even under extreme conditions, where the anomaly percentage is as low as 1%, *InoU* still delivers a robust performance.

H. Impacts of Contrastive and Adversarial Learning Techniques

We incorporate contrastive learning as a crucial part of the training of Siamese discriminator (Section III-D). Besides, adversarial learning is employed to further optimize the distribution of the manifold space and improve the expression of metrics related to discrepancies. In this section, we use experiments to compare the manifold space of the Siamese discriminator under different training settings.

Specifically, we visualize the manifold of the projection head $\mathcal{P}$ of the Siamese discriminator (as described in (32)) on the *SMAP* dataset using the *t-SNE* algorithm. During this comparison, three typical cases are considered:

- 1) Only With Cross Entropy. Train the Siamese discriminator using only (42).
- 2) Cross Entropy + Contrastive. Train the Siamese discriminator using both (41) and (44).
- 3) After Adversarial Tuning. Train the Siamese discriminator using both (41) and (44). Then, fine-tune the manifold with (45).

We can observe from Fig. 15 that the employment of the proposed training protocol optimizes the manifold space of the

Siamese discriminator and consequently enhances the detection performance.

For comparison, when only trained with cross entropy, the distribution of embeddings in the manifold appears more intermingled. As depicted in the lower dimension, the representation of samples, both normal and anomalous ones, are sparsely distributed in the manifold space. Consequently, the distinction between these two classes becomes blurred, which may compromise the detection performance when encountering unobserved or ambiguous anomalous samples if the samples are projected into overlapping regions.

On the other hand, the condensation is improved with the help of contrastive learning. This technique encourages the projection of samples to be clustered and informative. As a result, the compact localization of both normal and abnormal samples is achieved, thereby improving the clarity of the distribution characteristics.

In addition, the adversarial tuning further optimizes the manifold space of the Siamese discriminator. This process adjusts the distribution of the embeddings, so that the clusters become more pronounced and distinguishable. In this way, the separation between normal and anomalous samples is more distinct, which in turn enhances the detection capabilities of the Siamese discriminator.

V. RELATED WORK

The advancement of the industry has brought a surge in the scale of deployed assets. To ensure the smooth of their operation and reduce the cost of downtime, the reliability of these systems has become a critical concern.

Some works deploy their models near the deployed assets to get a reduced response time. Work [30] uses statistics *Ma-halanobis* distances as the metrics for early-stage fault diagnosis. Researchers [31] introduce clustering and k-means for fast anomaly filtering. However, performing anomaly detection locally on the device is challenging due to limited power and computation capacity. They then have to rely on a slim model or simple rules, resulting in spaces for performance improvement.

Other works embrace deep learning techniques in their workflow to get more robust results. Kim et al. [32] discuss the possibility of using *CNN* (Convolutional Neural Network) for feature extraction in anomaly detection. Canizo et al. [33] further adopt *CNN* and *RNN* (Recurrent Neural Network) as the backbone, using a classification-based workflow to achieve anomaly detection. However, discriminant models are prone to overfit when the dataset is not properly balanced, which is common in this scenario. Consequently, their performance gets worse in the presence of unobserved anomalous sequences. Considering this, researchers are shifting to generative models, i.e., *AE* (Autoencoder) and *VAE* (Variational Autoencoder). Luo et al. [34] propose an *AE* based anomaly detection approach for *WSN* (Wireless Sensor Network) that compares the observed and predicted sequences from sensors. Researchers in [9] combine *VAE* and *LSTM* to capture long-time dependencies in time series, so that short and long-period anomalies can be spotted. Su et al. [10] use planar normalizing to process

the stochastic variables in *VAEs*, and Li et al. [12] incorporate inter-metric and temporal embeddings to improve its anomaly detection performance on multivariate time series. *VAEs* provide a new perspective to reconstruct data, but their utilization is not straightforward. Being an unsupervised model, this model can hardly be trained when the distribution of the dataset is skewed. Besides, the design of *VAEs* needs careful consideration to avoid under-fit or over-fit problems [35].

Targeting the model over-expression problem, there are mainly two directions to rectify the reconstruction process. Some works incorporate memory modules to address the model over-expression problem [13], [15]. On the other hand, some works deploy an adversarial training protocol to force the decoder to learn a more robust representation translation approach [11]. Despite their designs, these solutions do not explicitly solve the problem of fine-grained discrepancy analysis, and training such structures is also challenging.

There are other endeavors to enhance the bridging and construction of dependencies in complex multivariate time series data. Inspired by the advancements in graph learning [36], [37], [38], [39], *GNNs* (Graph Neural Network), or more broadly, graph structures, are applied to enhance the overall performance. In Wu et al. [8], the authors investigate the application of *GNN* (Graph Neural Network) in IoT anomaly detection. Chen et al. [40] apply supervised contrastive learning to the graph structure, so that the potential anomalous nodes are identified and separated. Researchers in [41] use graphs to project subsequences of original data into embeddings and compare the characteristics. Ding et al. [42] use three graphs with inner-modal and inter-modal attentions to extract node dependencies in multi-modal IoT streams. Tuli et al. [27] involve a transformer in anomaly detection for better data interpretation. Xu et al. [28] further deploy the concept of transformer and adjacent-concentration to calculate association discrepancy. Yang et al. [43] propose a method to examine the correlation of representations in the latent space for anomaly detection. Despite the advancements in model design, these solutions face the challenge of huge overhead and scalability, and thus, may not be suitable for deployment in an industrial environment. Besides, most of these works depend on numerical discrepancies to identify anomalies, which may not be robust enough for complex industrial data.

VI. CONCLUSION

In this paper, we propose *InoU* as a scalable solution to the challenges in prior anomaly detection works. To enable the self-supervised training protocol and combat impacts from noise components in training materials, a *VAE* filter is employed in the processing pipeline of the target unlabeled data. We then develop a trio-attention U-Transformer model to capture intricate patterns and contextual information while maintaining outstanding scalability and efficiency. The ultra perception and intra-/ inter-flow attention based information aggregation enables the model to bridge dependencies across different levels with variable granularities, but without losing a global

perspective of the data. Furthermore, we design a Siamese discriminator that projects data into the manifold space, and adaptively collates discrepancies at the embedding level. We also introduce contrastive and adversarial learning to fine-tune the projection of manifold and boost the overall anomaly detection performance of *InoU* in industrial scenarios. Extensive experiments on five large-scale industrial datasets suggest significant performance improvements.

For future work, we propose investigating other deep learning methodologies, i.e., graph learning and meta learning, to enhance the construction of dependencies. We also plan to explore potential solutions for optimizing the training overheads.

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